

Conditional modal branch selection in physics-informed active control of lid-driven cavity flow: identification, reproducibility, and independent numerical verification

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Abstract

Active flow control of the lid-driven cavity is investigated using a parametric physics-informed neural network (PINN) combined with a truncated spatio-temporal Fourier actuation basis. Rather than assuming that optimization convergence identifies a unique optimal control, we quantify the conditional selection, reproducibility, and numerical realizability of low-dimensional control branches. Over Reynolds numbers $100 \leq Re \leq 1000$ and prescribed dimensionless actuation energies E^* , independent trainings reveal four empirical classes: B_1 (stationary second spatial harmonic), B_2 (time-dependent actuation), B_{first} (stationary first harmonic), and B_3 (mixed). Their frequencies depend on Re , E^* , initialization, loss weights, and control representation. Changing the variance-regularization weight redistributes branch accessibility without altering the Navier–Stokes problem; removing it selects the stationary first-harmonic branch B_{first} whereas the second harmonic prevails at the reference weight. Seed experiments at fixed (Re, E^*) yield distinct converged controls with similar losses. Selected controls are transferred without retraining to an independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver, validated against Ghia et al.; the second-harmonic control reduces integrated viscous dissipation by 76.5% at $Re = 500$ and 82.3% at $Re = 1000$ relative to a uniform lid, with grid-convergence analysis bounding the discretization uncertainty. A focused iso-energetic comparison shows that this frequently selected branch is actually more dissipative than a uniform lid at the same energy budget (dissipation ratio in $[1.072, 1.188]$), while the first harmonic is consistently less dissipative (ratio in $[0.646, 0.726]$), ranking first-harmonic < uniform < second-harmonic at both resolutions. Reproducibility experiments yield bit-identical double runs, and multi-seed audits confirm the branch taxonomy (0/185 reclassification disagreements); the results support conditional branch accessibility and reproducibility but refute uniqueness or global optimality.

Keywords: physics-informed neural networks; active flow control; lid-driven cavity; modal branch selection; lattice-Boltzmann method; reproducibility.

List of Abbreviations and Variables

1 INTRODUCTION

Active flow control seeks to modify fluid motion through externally imposed actuation in order to achieve a prescribed dynamical or energetic objective, including drag and lift modification, separation delay, mixing enhancement, and viscous dissipation reduction [1, 2]. The difficulty of this problem arises from the nonlinear and high-dimensional character of the Navier–Stokes

Symbol	Description
PINN	Physics-informed neural network
LBM	Lattice-Boltzmann method
MRT	Multiple-relaxation-time
DNS	Direct numerical simulation
PDE	Partial differential equation
D2Q9	Two-dimensional nine-velocity lattice
LOCO	Leave-one-Reynolds-condition-out
SR	Symbolic regression
$B_1, B_2, B_{\text{first}}, B_3$	Empirical branch classes
f_1	Stationary first-harmonic control-energy fraction
f_2	Stationary second-harmonic control-energy fraction
f_t	Explicitly time-dependent control-energy fraction
H_N	Normalized branch-selection entropy
Re	Reynolds number
E^*	Prescribed target actuation energy
E_r	Realized actuation energy
λ_{var}	Variance-regularization weight
AR	Cavity aspect ratio (L_x/L_y)
ε	Integrated viscous dissipation (LBM)
ω_{SR}	Symbolic-regression spatial frequency
R_K	Ratio of fluctuating to mean kinetic energy
GCI	Grid-convergence index
RMSE	Root-mean-square error

equations: even when the actuator is parameterized by a small number of degrees of freedom, its interaction with the flow produces multiple competing responses, strongly non-convex optimization landscapes, and, in many configurations, genuinely multiple stationary states. Reviews of flow control and machine-learning-based fluid mechanics have consequently emphasized the importance of reduced representations, interpretable flow structures, and systematic treatment of nonlinear effects rather than relying solely on scalar performance indicators [2, 1, 12, 13]. Methodological rigor in reporting convergence outcomes is especially important when physics-informed or data-driven methods are used, since a single converged solution may not represent the full landscape of accessible outcomes, and its physical significance can only be assessed by characterizing how robustly it is reached and whether it persists under independent numerical scrutiny.

Low-dimensional representations provide a natural route to tractable flow-control problems. Proper orthogonal decomposition, balanced truncation, dynamic mode decomposition, resolvent analysis, and related techniques have become standard tools for identifying dynamically important flow structures and for constructing reduced-order models amenable to analysis and control [3, 4, 15, 16]. In control applications, such representations substantially compress the optimization problem while retaining structures relevant to the dominant flow response. Reduced-order optimal-control formulations have been applied to direct numerical simulation (DNS)-based flow-control calculations [5, 6], and reinforcement-learning methods have extended this framework to complex geometries including bluff-body drag reduction [7, 8]. The lid-driven cavity occupies a canonical position in these investigations: its geometry and boundary conditions are simple, yet its recirculating flow structure and Reynolds-number dependence provide a non-trivial test of numerical and physical hypotheses, and high-quality benchmark solutions are available for independent verification [9]. At moderate and high Reynolds numbers the uncontrolled cavity admits multiple topological flow states, reinforcing the need for careful interpretation of any optimization outcome.

Physics-informed neural networks (PINNs) offer an alternative route to reduced representations [10, 11]. Rather than learning a flow response exclusively from precomputed simulation

data, PINNs incorporate the governing differential equations, boundary conditions, and physical constraints directly into the optimization objective through automatic differentiation of the network residuals. This formulation is particularly attractive for control identification because the flow field and the actuation parameters are optimized simultaneously, allowing the control itself to become an object of analysis rather than merely an output of a separate surrogate model. Machine learning has been applied broadly to fluid mechanics for model reduction, turbulence closure, and optimization [12, 13, 14], and physics-informed variants have demonstrated promising results for laminar, transitional, and turbulent flows. However, the physics-informed formulation does not remove the underlying non-convexity of the optimization problem. Different random initializations, network capacities, loss-weight configurations, regularization terms, or control parameterizations can lead to qualitatively different converged states. The repeated appearance of a particular low-dimensional structure cannot, by itself, establish that the structure is unique, globally optimal, or independent of the chosen representation.

This issue motivates a distinction that is rarely made explicit in physics-informed flow-control studies: the distinction between the *existence* of a controlled branch, its *accessibility* to a particular optimization procedure, and its *global optimality*. A control branch may be physically realizable and repeatedly recovered while remaining only one member of a set of competing converged states. Conversely, a branch that is infrequently recovered may not be physically less relevant; it may simply reside in a smaller basin of attraction under the adopted optimization formulation. The control parameterization introduces a further layer of conditioning: a finite basis does not merely reduce the number of degrees of freedom, but defines the subspace of control space in which the optimizer searches. An apparently low-dimensional control structure can therefore reflect both the dynamics of the controlled flow and the geometry of the chosen optimization space [4, 17, 20]. Subcritical non-convex landscapes in fluid mechanics have been characterized for variational and adjoint-based methods [18], and resolvent analysis has shown how structural assumptions can influence perceived optimal structures [19]. A quantitative framework for distinguishing these contributions is therefore essential before drawing physical conclusions from a repeatedly observed modal structure.

The present study addresses this branch-selection problem for distributed active boundary control of the two-dimensional lid-driven cavity. We formulate a physics-informed optimization in which the Navier–Stokes equations constrain the flow while a prescribed actuation-energy level and a dissipation-related objective determine the control. The actuation is represented in a truncated spatio-temporal Fourier basis, and a parametric neural representation identifies controls over the investigated Reynolds-number interval. Rather than treating the dominant modal structure as an assumed optimum, we explicitly test its recoverability, multiplicity, and parameter dependence. Independent training realizations estimate empirical branch-selection frequencies

$$\hat{P}(B_i \mid Re, E^*, \lambda_{\text{var}}), \quad (1)$$

while systematic variations of Re , E^* , network capacity, control-space truncation, cavity aspect ratio, loss weighting, and control basis determine when the observed branches persist or change. The resulting branch structure is then subjected to an independent numerical test: selected controls are transferred without retraining to an independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver, providing a completely independent discretization and implementation. Grid refinement and benchmark comparisons quantify the associated numerical uncertainty, and symbolic regression provides a compact analytical approximation enabling assessment of cross-condition generalization. The objective is not to claim the discovery of a new universal cavity mode or to establish global optimality, but to provide a quantitative framework for distinguishing branch existence, branch accessibility, and branch reproducibility in physics-informed flow-control optimization.

The central questions of this study are therefore: (i) Does a distributed physics-informed control problem repeatedly select a low-dimensional modal branch over a range of flow and ac-

tuation parameters? (ii) Do independent initializations reveal competing branches with comparable optimization performance? (iii) How does the variational structure of the loss redistribute the empirical accessibility of those branches? (iv) Does the observed modal structure survive changes in the control representation? (v) Can the resulting branches be independently realized and distinguished by a conventional numerical flow solver? Answering these questions provides a more stringent interpretation of low-dimensional structures obtained from physics-informed optimization and places modal concentration within the broader framework of nonlinear branch selection in controlled fluid flows. The remainder of the paper is organized as follows. Section 2 formulates the control problem and PINN framework. Section 3 presents the global branch landscape. Section 4 investigates conditional accessibility. Section 5 examines robustness to capacity, dimension, geometry, and basis. Section 6 presents the independent LBM verification. Section 7 synthesizes the results. Section 8 presents the symbolic representation. Section 9 states the conclusions.

2 PROBLEM FORMULATION AND PHYSICS-INFORMED OPTIMIZATION

The objective is to examine how a distributed boundary-control problem can admit competing low-dimensional actuation branches and how their accessibility changes with the physical and variational parameters of the identification procedure. Before introducing the physics-informed optimization, the controlled flow problem and the admissible actuation space must be specified independently of the neural network representation. All notation is consistent throughout the paper; additional implementation details are provided in the Supplementary Material.

2.1 Controlled cavity and governing equations

We consider a two-dimensional incompressible lid-driven cavity in which the upper boundary is subjected to a spatially and temporally distributed tangential actuation. The lower and lateral walls remain stationary with no-slip conditions. The lid velocity is not prescribed *a priori* as a uniform profile but is treated as a control variable drawn from a finite-dimensional admissible space. This is essential for the subsequent branch analysis: the branches investigated here are not solutions of different physical problems, but different control-flow realizations admitted by the same governing equations, boundary geometry, and actuation constraints.

After nondimensionalization by the cavity height L_y^{dim} , the wall-normal extent is unity and the streamwise extent equals the aspect ratio $AR = L_x^{\text{dim}}/L_y^{\text{dim}}$:

$$0 \leq x \leq L_x, \quad 0 \leq y \leq 1, \quad L_x = AR. \quad (2)$$

The reference square cavity corresponds to $L_x = 1$. Velocities are nondimensionalized by a reference lid velocity U_0 , time by L_y^{dim}/U_0 , and pressure by ρU_0^2 . The Reynolds number is

$$Re = \frac{U_0 L_y^{\text{dim}}}{\nu}, \quad (3)$$

where ν is the kinematic viscosity. Note that U_0 is a scaling velocity, not a requirement that the lid remain uniform; the distributed control U_{lid} may change sign locally provided it remains within the admissible control space and satisfies the prescribed energy constraint.

The controlled velocity field $\mathbf{u} = (u, v)$ and kinematic pressure p satisfy the dimensionless incompressible Navier–Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \frac{1}{Re} \nabla^2 \mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0. \quad (4)$$

The stationary boundaries satisfy the no-slip condition

$$\mathbf{u} = \mathbf{0}, \quad x = 0, x = L_x, y = 0, \quad (5)$$

and the controlled upper boundary satisfies

$$u(x, 1, t; Re) = U_{\text{lid}}(x, t; Re), \quad v(x, 1, t; Re) = 0. \quad (6)$$

The problem is nonlinear for two reasons: the Navier–Stokes convective term, and the boundary condition (6), which is itself selected from a parameterized control space. The optimization therefore couples the selection of a boundary actuation to the nonlinear flow response that the same actuation generates.

2.2 Finite-dimensional spatio-temporal control

The lid actuation is represented in a truncated spatio-temporal Fourier basis. The sine basis in x enforces $U_{\text{lid}}(0, t; Re) = U_{\text{lid}}(L_x, t; Re) = 0$, preventing discontinuous tangential velocity at the upper corners for every tested aspect ratio. The cosine basis in t allows the $n = 0$ terms to represent purely stationary actuation, the most directly interpretable patterns physically. The lid velocity is expanded as

$$U_{\text{lid}}(x, t, Re) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t-1} a_{mn}(Re) \sin\left(\frac{m\pi x}{L_x}\right) \cos(n\pi t). \quad (7)$$

All modal labels use the physical (one-based) convention: the dominant stationary second spatial harmonic corresponds to $(m, n) = (2, 0)$. The baseline control space uses $(N_x, N_t) = (6, 5)$, giving 30 nominal coefficients. Control-space sensitivity examines

$$(N_x, N_t) \in \{(4, 3), (6, 1), (6, 3), (6, 5), (8, 5), (8, 9)\}, \quad (8)$$

spanning 12 to 72 nominal modes. The modal energy of coefficient (m, n) is

$$\mathcal{E}_{m0} = \frac{1}{2}a_{m0}^2, \quad \mathcal{E}_{mn} = \frac{1}{4}a_{mn}^2 \quad (n \geq 1), \quad (9)$$

where the factors account for temporal basis normalization ($\langle \cos^2(0) \rangle = 1$, $\langle \cos^2(n\pi t) \rangle = \frac{1}{2}$ for $n \geq 1$). The normalized modal fraction is

$$f_{mn} = \frac{\mathcal{E}_{mn}}{\sum_{i=1}^{N_x} \sum_{j=0}^{N_t-1} \mathcal{E}_{ij}}, \quad \sum_{m=1}^{N_x} \sum_{n=0}^{N_t-1} f_{mn} = 1. \quad (10)$$

The stationary second-harmonic fraction is $f_2 \equiv f_{20}$, and the total time-dependent fraction is $f_t = \sum_{m=1}^{N_x} \sum_{n=1}^{N_t-1} f_{mn}$. The realized actuation energy is

$$E_r = \langle U_{\text{lid}}^2 \rangle_{s,t} = \frac{1}{T} \int_0^T \int_0^1 U_{\text{lid}}^2(s, t; Re) ds dt, \quad s = \frac{x}{L_x}, \quad (11)$$

and the optimization is conditioned on $E_r \approx E^*$. Modal-energy quadrature conventions are detailed in Supplementary Sec. S2.

2.3 Parametric PINN formulation

The flow field and Reynolds-dependent control coefficients are represented simultaneously by a parametric neural network with inputs $(x, y, t, \widetilde{Re})$, where

$$\widetilde{Re} = \frac{Re - 550}{450} \in [-1, 1], \quad (12)$$

maps the training interval $[100, 1000]$ to $[-1, 1]$. A single trained network represents the flow-control pair over the entire Reynolds-number range, enabling efficient evaluation of the branch map without repeated retraining at each condition. The flow variables (u, v, p) and control coefficients $\{a_{mn}(\widetilde{Re})\}$ are the network outputs. Two subnetworks are employed: a flow network with 6 hidden layers of 64 neurons each using tanh activation (21 379 trainable parameters), and a control network with 4 hidden layers of 32 neurons each (3 614 parameters). All computations are performed in double-precision arithmetic, and all physical residuals are evaluated by automatic differentiation. Full architecture specifications are provided in Supplementary Sec. S1.

2.4 Optimization objective and training procedure

The multi-objective loss combines partial-differential-equation (PDE), boundary-condition, dissipation, control-energy, variance, and Reynolds-separation terms:

$$\mathcal{L} = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{L}_{\text{BC}} + \lambda_{\text{diss}} \mathcal{L}_{\text{diss}} + \lambda_{\text{ctrl}} \mathcal{L}_{\text{ctrl}} + \lambda_{\text{var}} \mathcal{L}_{\text{var}} + \lambda_{Re} \mathcal{L}_{Re}, \quad (13)$$

with reference weights $\lambda_{\text{PDE}} = 1$, $\lambda_{\text{BC}} = 15$, $\lambda_{\text{diss}} = 0.05$, $\lambda_{\text{ctrl}} = 200$, $\lambda_{\text{var}} = 1$, $\lambda_{Re} = 2$. The individual loss components are:

PDE residual loss: $\mathcal{L}_{\text{PDE}} = \langle R_u^2 + R_v^2 + R_c^2 \rangle_{\mathcal{X}_{\text{int}}}$, where R_u, R_v, R_c are momentum and continuity residuals evaluated at $N_{\text{phys}} = 4000$ interior collocation points resampled uniformly at each epoch.

Boundary condition loss: $\mathcal{L}_{\text{BC}} = \langle u^2 + v^2 \rangle_{\Gamma_{\text{stat}}} + \langle [u - U_{\text{lid}}]^2 + v^2 \rangle_{\Gamma_{\text{top}}}$, where $\Gamma_{\text{stat}} = \{x = 0\} \cup \{x = L_x\} \cup \{y = 0\}$. The weight $\lambda_{\text{BC}} = 15$ enforces boundary conditions strongly relative to the interior residual.

Viscous dissipation objective:

$$\mathcal{L}_{\text{diss}} = \int_{\Omega} \frac{2}{Re} [u_x^2 + v_y^2 + \frac{1}{2}(u_y + v_x)^2] d\Omega. \quad (14)$$

This term drives the optimizer toward low-dissipation actuation patterns and constitutes the physical objective of the control problem.

Actuation energy penalty: $\mathcal{L}_{\text{ctrl}} = (E_r - E^*)^2$. The large weight $\lambda_{\text{ctrl}} = 200$ enforces the energy budget tightly, ensuring that competing converged solutions are compared at the same actuation energy level and that the modal competition is not trivially explained by different energy budgets.

Variance regularization:

$$\mathcal{L}_{\text{var}} = -\log[\text{Var}(U_{\text{lid}}) + 10^{-5}]. \quad (15)$$

This term penalizes spatially uniform or near-uniform lid profiles by rewarding higher spatial variance. It plays a critical role in breaking degeneracy toward trivially flat controls and is the primary determinant of branch selection, as demonstrated in Section 4.

Reynolds-separation regularization:

$$\mathcal{L}_{Re} = \max \left[0, \delta - \frac{1}{N_{\text{probe}}} \sum_{i=1}^{N_{\text{probe}}} (U_{\text{lid}}(x_i, t_0, Re_{\text{min}}) - U_{\text{lid}}(x_i, t_0, Re_{\text{max}}))^2 \right], \quad (16)$$

with $N_{\text{probe}} = 20$, $t_0 = 0.5$, $\delta = 0.01$, preventing the parametric network from collapsing to a Reynolds-independent control.

Training consists of a 300-epoch pre-training phase in which only the flow network is updated with control coefficients frozen at zero, establishing an initial physically consistent flow representation. A 3000-epoch joint optimization phase follows using the Adam optimizer with initial learning rate $\eta_0 = 8 \times 10^{-4}$ and exponential decay by factor 0.995 per 100 epochs. Collocation points are resampled at every epoch. Complete sampling details are provided in Supplementary Sec. S1.

2.5 Branch classification and reproducibility protocol

Converged neural realizations are classified into four empirical branch classes according to their modal signature computed from Eqs. (9)–(10). Let $f_1 \equiv f_{10}$ denote the stationary first-harmonic (fundamental) fraction, $f_2 \equiv f_{20}$ the stationary second-harmonic fraction, and $f_t = \sum_m \sum_{n \geq 1} f_{mn}$ the total explicitly time-dependent fraction:

- **Branch B_{first} — Stationary first-harmonic dominated:** $f_1 > 0.5$. Physical harmonic $(m, n) = (1, 0)$.
- **Branch B_1 — Stationary second-mode dominated:** $f_2 > 2f_t$ and $f_2 > 0.5$. Physical harmonic $(m, n) = (2, 0)$. Ensemble mean $\overline{f_2} = 85.3\%$, $\overline{f_t} = 7.4\%$.
- **Branch B_2 — Temporally dominated:** $f_t > 2f_2$ and $f_t > 0.5$. Physical harmonic $(m, n) = (1, 1)$. Ensemble mean $\overline{f_2} = 7.5\%$, $\overline{f_t} = 88.1\%$.
- **Branch B_3 — Mixed:** none of the above. Ensemble mean $\overline{f_2} = 34.8\%$, $\overline{f_t} = 43.9\%$.

The empirical selection frequency over N_s independent seeds is $\hat{P}(B_k) = N_k/N_s$. Selection diversity is measured by normalized Shannon entropy $H_N = -\frac{1}{\ln 4} \sum_k \hat{P}(B_k) \ln \hat{P}(B_k)$, which equals zero for deterministic single-branch selection and unity for a uniform distribution over all four classes. The reference campaign of Sec. 3, conducted at $\lambda_{\text{var}} = 1$, contains no B_{first} realization; this branch becomes populated only when the variance regularizer is weakened or removed (Sec. 4.1). In the original three-class scheme these first-harmonic realizations were grouped with the mixed branch B_3 ; the explicit four-class classification shows that they constitute a distinct attractor. Detailed implementation parameters and realization-level campaign data are provided in the Supplementary Material.

3 GLOBAL BRANCH-SELECTION LANDSCAPE

3.1 Broad (Re, E^*) campaign

The broad campaign evaluated 185 independent training runs across 30 parameter-space cells ($Re \in \{100, 300, 500, 700, 900, 1000\}$, $E^* \in \{0.1, 0.25, 0.5, 1, 2\}$), with at least 5 seeds per cell and up to 20 at selected cells of particular interest. Figure 1 presents the resulting empirical branch-frequency map $\hat{P}(B_k)$, and Table 1 summarizes the global realization statistics. The complete realization-level dataset is provided in Supplementary Sec. S3 and Supplementary Table S1.

Across the 185 runs, B_1 , B_2 , and B_3 account for 42.2%, 43.8%, and 14.1% of solutions, respectively. The near balance between B_1 and B_2 is the most significant feature of the global landscape: while the stationary second-harmonic branch is frequently accessible, it does not dominate the ensemble to the point of being the unique or near-deterministic outcome. The mixed branch B_3 is the least frequently selected, arising primarily at elevated actuation energies $E^* \geq 1$ and higher Reynolds numbers. No B_{first} realization occurs anywhere in this campaign: the first-harmonic fraction remains below $f_1 = 0.13$ throughout, confirming that the reference

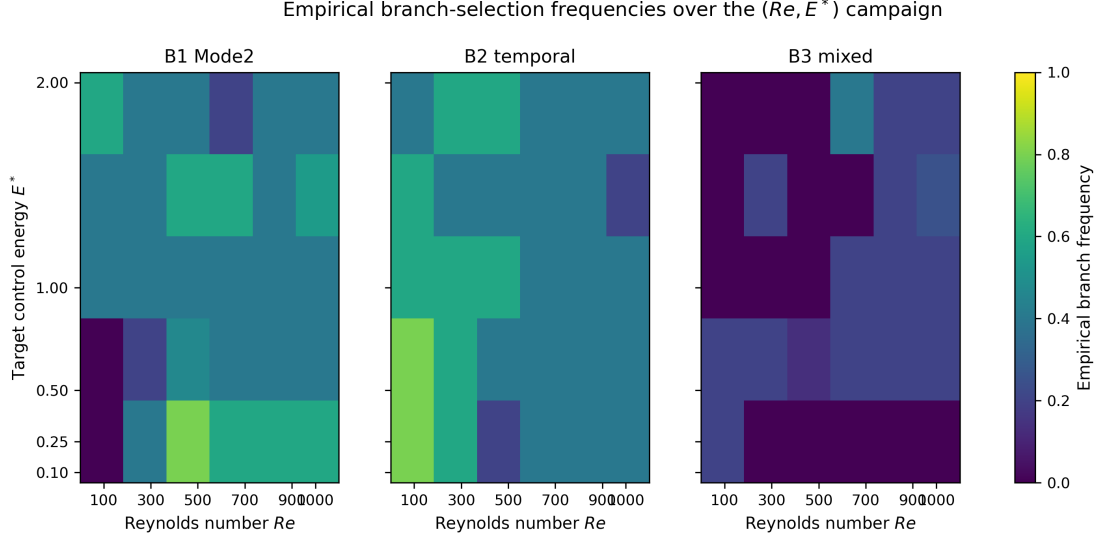


Figure 1: Empirical branch-selection frequencies over the (Re, E^*) parameter space from the 185-realization campaign. Each panel reports the fraction of converged realizations classified as B_1 (stationary second-harmonic dominated), B_2 (temporally dominated), or B_3 (mixed) at each parameter cell. The values represent empirical recovery frequencies from independent optimizations and should not be interpreted as deterministic phase boundaries. Cell-level realization counts and realization-level classifications are reported in Supplementary Table S1.

Table 1: Global statistics of the empirical branch classes populated over the 185-realization (Re, E^*) campaign at $\lambda_{\text{var}} = 1$ (no B_{first} realization occurs at this weight). Modal fractions in percentages.

Branch	N	Frequency (%)	$\overline{f_2}$ (%)	Median f_2 (%)	$\overline{f_t}$ (%)	Median f_t (%)
B_1	78	42.2	85.3	88.9	7.4	3.8
B_2	81	43.8	7.5	3.5	88.1	95.4
B_3	26	14.1	34.8	36.3	43.9	45.1

weight $\lambda_{\text{var}} = 1$ actively excludes the fundamental mode. The branch frequencies exhibit a clear Re -dependence: B_1 is more readily accessible at moderate Re around 500, while at $Re \in [100, 300]$, B_2 often dominates. This suggests that the relative depths of the two competing optimization basins vary systematically with the flow regime.

3.2 Continuous modal structure in observable space

Figure 2 illustrates the distribution of all 185 realizations in the two-dimensional (f_2, f_t) observable space. Three distinct clusters are clearly visible, separated by low-density gaps. The B_1 cluster concentrates near the $f_t \approx 0$ axis at high f_2 ; the B_2 cluster concentrates near the $f_2 \approx 0$ axis at high f_t ; and the B_3 cluster occupies an intermediate region. The cluster separation establishes that the branch classes are well-defined structural entities rather than arbitrary divisions of a continuous distribution. The first-harmonic class B_{first} does not appear in this observable space because it is not populated at $\lambda_{\text{var}} = 1$; it emerges only for weakened variance regularization (Sec. 4.1).

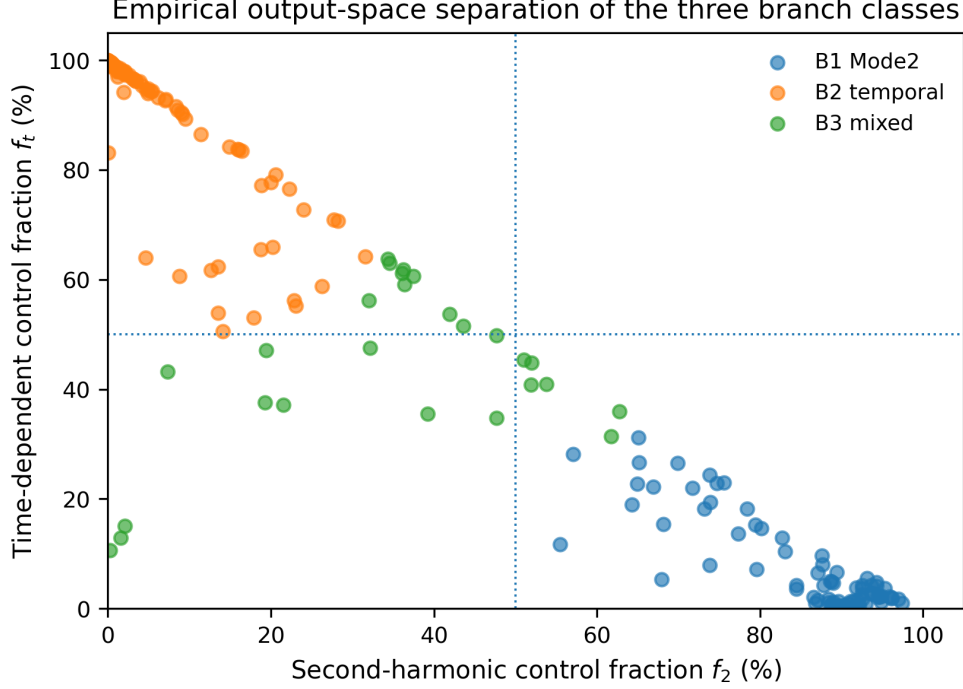


Figure 2: Distribution of the 185 converged realizations in the (f_2, f_t) modal observable space, colored by assigned empirical branch class (B_1 : red; B_2 : blue; B_3 : green). Three well-separated clusters confirm that the empirical branch classes correspond to structurally distinct attractors of the physics-informed optimization.

3.3 Conditional branch accessibility and non-determinism

A key observation from Figure 1 is that (Re, E^*) does not uniquely determine a single control class. At $Re = 500$, $E^* = 0.25$, a 15-seed sub-campaign yields $\hat{P}(B_1) = 0.467$, $\hat{P}(B_2) = 0.400$, $\hat{P}(B_3) = 0.133$, demonstrating genuine multi-branch accessibility at a single operating point. At $Re = 1000$, $E^* = 1$, a 20-seed sub-campaign yields $\hat{P}(B_1) = 0.55$, $\hat{P}(B_2) = 0.20$, $\hat{P}(B_3) = 0.25$. The mapping $(Re, E^*) \rightarrow \mathcal{B}$ is therefore set-valued: each operating point maps to an empirical distribution $\{\hat{P}(B_1), \hat{P}(B_2), \hat{P}(B_3)\}$. The following sections investigate the conditions under which this distribution changes.

Finally, the temporal structure of converged realizations was examined at the trajectory level for seeds representative of the three classes (seed 7 $\rightarrow B_1$, seed 8 $\rightarrow B_2$, seed 4 $\rightarrow B_3$). A clear four-epoch spectral peak in the raw per-epoch training loss (period ≈ 4 epochs; power up to $\sim 55\times$ the spectral median) does *not* propagate to the control energy, the control coefficients, or the modal fractions: no period-4 signature is found in the control path at the sampling scale, and the branch labels remain stable across transients. The four-epoch motif is therefore an optimizer-internal oscillation confined to the loss trajectory, not a physical or control-space cycle. The three classes differ in their established behavior: the B_1 control is effectively frozen (control displacement $D_1 \simeq 0.009$), the B_2 realization shows a finite relocation transient after which the control freezes (stationary $D_1 \simeq 0.008$), and the B_3 realization exhibits a slow, persistent drift of both the control coefficients and the modal fractions (stationary modal displacement $\Delta f(4) \simeq 0.05$, about $50\times$ the B_1 value). Autocorrelation functions, cycle tests, and their definitions are provided in Supplementary Sec. S11.

Table 2: Empirical branch-selection frequencies from the λ_{var} sensitivity campaign. Each entry is based on five independent realizations at fixed (Re, E^*) . H_N is the normalized branch-selection entropy (normalised by $\ln 4$). Complete 95% confidence intervals are provided in Supplementary Sec. S4.

Re	E^*	λ_{var}	$\hat{P}(B_1)$	$\hat{P}(B_2)$	$\hat{P}(B_{\text{first}})$	$\hat{P}(B_3)$	H_N	Dominant class
500	0.25	0	0.0	0.0	1.0	0.0	0.000	B_{first}
500	0.25	0.25	0.6	0.4	0.0	0.0	0.485	B_1
500	0.25	0.5	0.4	0.2	0.0	0.4	0.761	B_1/B_3
500	0.25	1	0.4	0.2	0.0	0.4	0.761	B_1/B_3
500	0.25	2	0.4	0.4	0.0	0.2	0.761	B_1/B_2
500	0.25	5	0.4	0.6	0.0	0.0	0.485	B_2
700	2	0	0.0	0.0	1.0	0.0	0.000	B_{first}
700	2	0.25	0.0	0.0	1.0	0.0	0.000	B_{first}
700	2	0.5	0.0	0.6	0.0	0.4	0.485	B_2
700	2	1	0.2	0.4	0.0	0.4	0.761	B_2/B_3
700	2	2	0.6	0.2	0.0	0.2	0.685	B_1
700	2	5	0.6	0.2	0.0	0.2	0.685	B_1
1000	1	0	0.0	0.0	1.0	0.0	0.000	B_{first}
1000	1	0.25	0.0	0.6	0.0	0.4	0.485	B_2
1000	1	0.5	0.4	0.4	0.0	0.2	0.761	B_1/B_2
1000	1	1	0.4	0.4	0.0	0.2	0.761	B_1/B_2
1000	1	2	0.4	0.4	0.0	0.2	0.761	B_1/B_2
1000	1	5	0.4	0.6	0.0	0.0	0.485	B_2

4 CONDITIONAL BRANCH ACCESSIBILITY

4.1 Variational dependence on λ_{var}

The effect of variance regularization was tested in a dedicated 90-run campaign ($N_s = 5$ seeds at 3 operating points, 6 values of $\lambda_{\text{var}} \in \{0, 0.25, 0.5, 1, 2, 5\}$). Table 2 and Figure 3 demonstrate that changing λ_{var} systematically redistributes empirical branch-selection frequencies while leaving the Navier–Stokes problem entirely unchanged. Complete realization-level results and finite-sample Wilson-score confidence intervals are given in Supplementary Sec. S4.

At $\lambda_{\text{var}} = 0$, all three operating conditions yield exclusively B_{first} realizations ($H_N = 0$): in the absence of spatial-variance regularization the optimizer collapses onto the lowest-order stationary mode, the single-lobe fundamental $\sin(\pi x)$, which is spatially structured rather than mixed. Introducing any nonzero λ_{var} immediately suppresses B_{first} and shifts accessibility toward B_1 or B_2 depending on the operating point. At $(Re, E^*) = (700, 2)$ the first-harmonic branch persists up to $\lambda_{\text{var}} = 0.25$ and disappears only for larger values, whereas at the other two operating points it is removed by the smallest tested nonzero weight; the transition between B_{first} and the second-order branches is therefore sharp in λ_{var} . For $\lambda_{\text{var}} \geq 0.5$ the distribution redistributes non-monotonically: at $(Re, E^*) = (500, 0.25)$, B_1 frequency is maximized at intermediate $\lambda_{\text{var}} \approx 0.25$, whereas larger values shift the distribution toward B_2 , reaching $\hat{P}(B_2) = 0.6$ at $\lambda_{\text{var}} = 5$. At $(Re, E^*) = (700, 2)$, the opposite trajectory is observed: values in the range $[2, 5]$ favor B_1 even though small values favor B_2 or B_3 . These results demonstrate that variance regularization acts as an optimization landscape bias: it modifies which basin of attraction the optimizer enters during training without modifying the underlying fluid-mechanics problem. This constitutes direct evidence that the branch structure observed in physics-informed control studies is at least partially a property of the variational formulation rather than solely of the physical problem.

4.2 Dependence on actuation energy

Target-energy sweeps at fixed $Re = 500$ over $E^* \in \{0.01, 0.05, 0.1, 0.25, 0.5, 1, 2\}$ reveal that the second-harmonic purity f_2 decreases monotonically as the prescribed actuation amplitude grows,

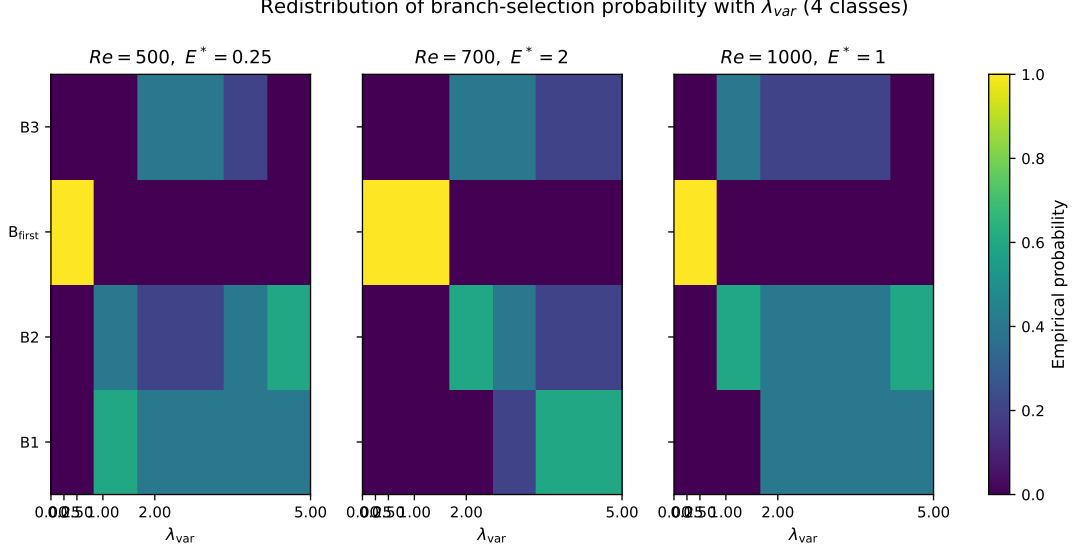


Figure 3: Empirical redistribution of branch-selection frequencies with the variance-regularization weight λ_{var} for three fixed operating conditions. Heatmaps show the four classes B_1 , B_2 , B_{first} , and B_3 ; cell values are empirical frequencies estimated from five independent realizations per λ_{var} value. 95% Wilson-score confidence intervals are tabulated in Supplementary Sec. S4.

while f_t correspondingly increases (Figure 4). At the lowest energies ($E^* \leq 0.05$), the control budget is insufficient to explore temporal degrees of freedom, and the optimizer concentrates almost entirely on the second spatial harmonic ($f_2 \approx 93\text{--}94\%$). At the reference condition $E^* = 0.25$, $f_2 = 92.2\%$ with $f_t = 1.3\%$. At $E^* = 2$, however, f_2 drops to 74.0% while f_t rises to 21.8% , indicating that at large actuation amplitudes the optimizer increasingly distributes control energy away from the second harmonic toward additional spatial and temporal components. This transition reflects the fact that at high energy the energy budget can be satisfied by many different modal mixtures, making the second-harmonic basin less uniquely attractive. Throughout the sweep the first-harmonic fraction remains small ($f_1 \leq 8\%$), so the B_{first} branch is never selected at the reference weight. Extended energy sweep datasets are provided in Supplementary Sec. S5.

4.3 Initialization sensitivity

Ten independent random seeds at $Re = 500$, $E^* = 0.25$ produced 4 second-mode dominated (B_1), 4 temporally dominated (B_2), and 2 intermediate (B_3) realizations. Figure 5 illustrates the seed-to-seed outcomes in the (f_2, f_t) space. No seed selects the first-harmonic branch ($f_1 < 0.02$ for all ten seeds), confirming that B_{first} is not a low-energy accidental outcome at the reference weight. Despite qualitatively distinct modal structures, total training losses are tightly clustered within $[1.371, 1.394]$, a relative spread of only 1.7% . The B_1 and B_2 realizations achieve essentially equivalent loss values, confirming that the two branches are separated by distinct basins of comparable depth rather than by a large energy gap. This multiplicity is the direct signature of a non-convex optimization problem with competing local minima, and it demonstrates that the selection of a particular branch by any single training run cannot serve as evidence of global optimality. The complete seed-level dataset is provided in Supplementary Sec. S5.

Machine-level reproducibility was verified independently for each of the three branch classes through idempotent double runs at the reference condition $Re = 500$, $E^* = 0.25$ using seeds representative of B_1 , B_2 , and B_3 (seeds 7, 8, and 4, respectively). The two runs of each pair are

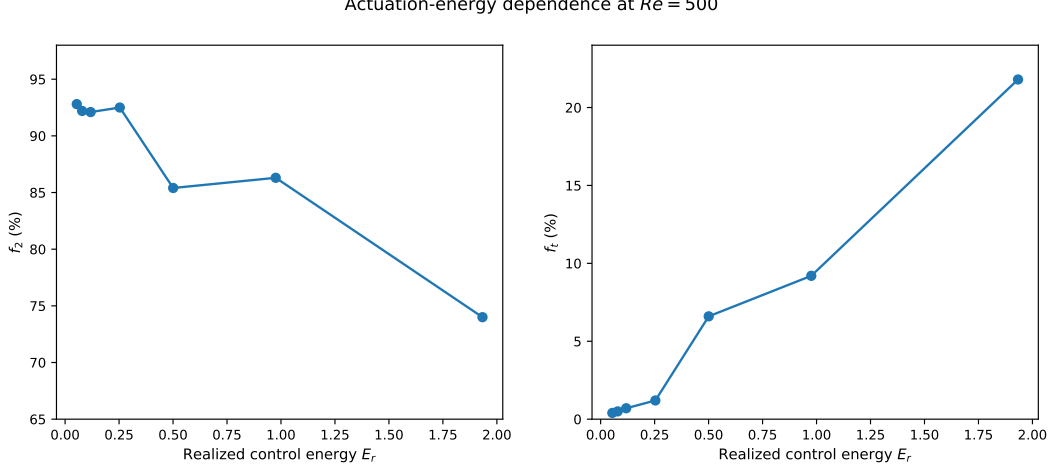


Figure 4: Dependence of modal structure on prescribed actuation energy E^* at $Re = 500$. Left: second-harmonic fraction f_2 and time-dependent fraction f_t across the energy sweep. Right: second-harmonic amplitude A_2 and realized energy E_r versus E^* . Second-harmonic dominance is robust at low to moderate energies but erodes at $E^* \geq 1$.

Table 3: Loss-term ablation results at $Re = 500$, $E^* = 0.25$. The first-harmonic fraction f_1 isolates the modal-class shift induced by removing $\mathcal{L}_{\text{var}}$.

Configuration	E_r	f_1 (%)	f_2 (%)	f_t (%)	Class
Baseline	0.2534	0.0	92.2	1.3	B_1
No $\mathcal{L}_{\text{var}}$	0.2582	90.4	0.2	5.1	B_{first}
No $\mathcal{L}_{Re}$	0.2534	0.0	92.2	1.3	B_1
No $\mathcal{L}_{\text{diss}}$	0.2533	0.0	92.3	1.2	B_1
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{Re}$	0.2582	90.4	0.2	5.1	B_{first}
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{\text{diss}}$	0.2578	90.5	0.3	5.0	B_{first}
$\lambda_{\text{var}}/2$	0.2546	0.2	89.3	1.5	B_1
$2\lambda_{\text{var}}$	0.2598	0.1	92.8	1.0	B_1

bit-identical to the last decimal for seeds 7 and 8 (zero difference in f_2 , f_t , and total loss) and quasi-identical for seed 4 (branch class B_3 unchanged; $|\Delta f_2| = 0.012$), and each pair coincides, after re-analysis with the final code, with the original study realization of the same seed. The reproducibility of the branch taxonomy is therefore established per branch class rather than only through aggregate frequencies. Detailed tables are provided in Supplementary Sec. S10.

4.4 Loss-term ablation

Ablation experiments at $Re = 500$, $E^* = 0.25$ isolate the role of individual loss components. Table 3 reports results for single and dual term removals. Removing $\mathcal{L}_{\text{var}}$ alone causes f_2 to collapse from 92.2% to 0.2% while the first-harmonic fraction f_1 jumps from 0.0% to 90.4%: the modal class shifts completely from the second-harmonic branch B_1 to the first-harmonic branch B_{first} . Removing $\mathcal{L}_{Re}$ or $\mathcal{L}_{\text{diss}}$ individually leaves f_2 essentially unchanged at 92.2–92.3%. Dual ablations confirm that the shift induced by removing $\mathcal{L}_{\text{var}}$ is not amplified or modulated by simultaneously removing other terms; in every case the outcome is B_{first} . Reducing λ_{var} by half decreases second-mode concentration modestly to 89.3%, while doubling it increases it to 92.8%. These results establish that $\mathcal{L}_{\text{var}}$ is the critical loss term for B_1 selection and, equivalently, for excluding B_{first} , and that its action is not redundant with the dissipation or Re -separation terms. Additional ablation datasets are given in Supplementary Sec. S5.

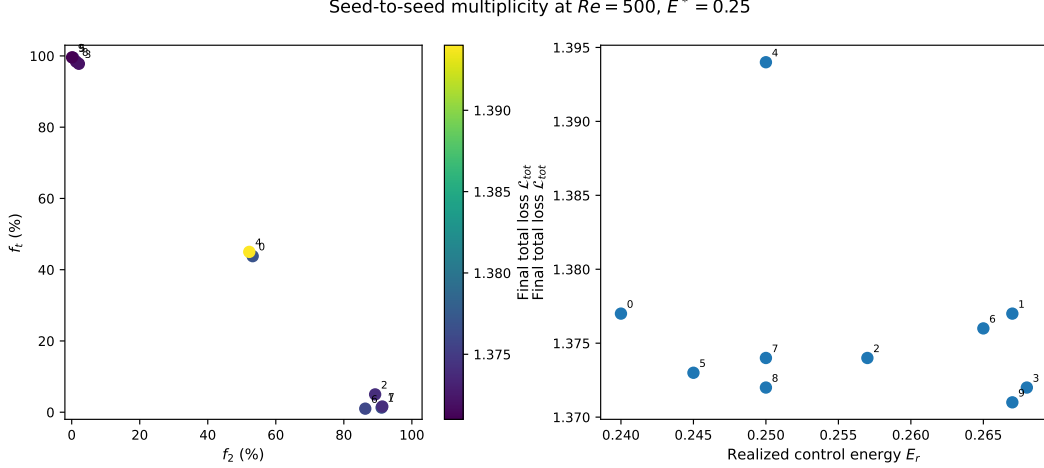


Figure 5: Seed-to-seed multiplicity at $Re = 500$, $E^* = 0.25$ across 10 independent initializations. Left: modal outcome in the (f_2, f_t) plane, labeled by seed index and colored by total training loss. Right: realized control energy E_r versus total loss. Total losses lie within $[1.371, 1.394]$, confirming that B_1 and B_2 correspond to competing basins of comparable depth.

Table 4: Control-truncation sensitivity at $Re = 500$, $E^* = 0.25$.

(N_x, N_t)	Modes	A_2	E_r	f_2 (%)	f_t (%)
(4,3)	12	0.671	0.2541	87.6	2.1
(6,1)	6	0.678	0.2533	89.2	0.0
(6,3)	18	0.678	0.2778	89.3	1.8
(6,5)	30	0.684	0.2533	92.3	1.3
(8,5)	40	0.669	0.2475	87.8	2.0
(8,9)	72	0.668	0.2775	87.3	2.4

5 ROBUSTNESS AND REPRESENTATION DEPENDENCE

5.1 Neural-network capacity

To assess whether the second-harmonic concentration is an artifact of the specific network architecture, three capacity variants were evaluated at $Re = 500$, $E^* = 0.25$: a small network (4×32), the reference network (6×64), and a large network (8×128). The second-harmonic fraction remains in the range $f_2 \in [89.5\%, 92.2\%]$ across all three configurations, a spread of only 2.7%. The fact that a network roughly 40 times smaller than the large variant produces essentially the same modal concentration argues against the hypothesis that the second-harmonic structure is imposed by the inductive bias of the neural network. Detailed architecture-level results are reported in Supplementary Sec. S6.

5.2 Control-space dimension

Truncation sweeps across six control dimensions (Table 4) show persistent second-mode dominance with $f_2 \in [87.3\%, 92.3\%]$. Figure 6 confirms that expanding the basis to 72 modes does not eliminate the B_1 branch or substantially redistribute modal energy toward higher harmonics; additional modes remain largely unactivated. The second-harmonic amplitude A_2 varies only from 0.668 to 0.684 across all truncations. These results support the hypothesis that the second-harmonic structure reflects a genuine low-dissipation response of the cavity flow rather than a numerical artifact of insufficient degrees of freedom. Truncation datasets are provided in Supplementary Sec. S6.

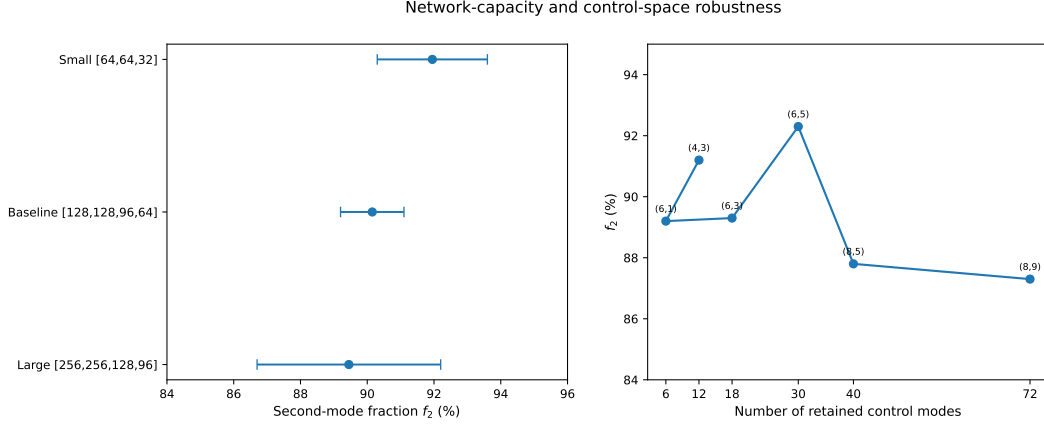


Figure 6: Control-truncation sensitivity at $Re = 500$, $E^* = 0.25$. Left: f_2 and f_t across six tested truncations ordered by total nominal modes. Right: second-harmonic amplitude A_2 and realized energy E_r . The B_1 branch persists with $f_2 \geq 87.3\%$ across all tested dimensions.

Table 5: Aspect-ratio dependence of the modal structure at $Re = 500$, $E^* = 0.25$.

AR	A_2	E_r	f_2 (%)	f_t (%)
0.5	0.6034	0.2603	69.9	26.7
1.0	0.6835	0.2534	92.2	1.3
2.0	0.6770	0.2514	91.2	1.6

5.3 Cavity aspect ratio

Changing the cavity geometry ($AR \in \{0.5, 1, 2\}$) at $Re = 500$, $E^* = 0.25$ shows that the stationary second-harmonic signature remains dominant at the reference aspect ratio ($f_2 = 92.2\%$) and at the elongated geometry ($AR = 2$, $f_2 = 91.2\%$), but its concentration decreases notably at $AR = 0.5$, where f_2 drops to 69.9% while f_t rises to 26.7% (Table 5). The compressed geometry changes the relationship between the harmonic structure of the Fourier basis and the natural recirculation scales of the cavity: at $AR = 0.5$, the second harmonic has a half-wavelength of 0.25, shorter than the natural recirculation scale, and may be less efficient at redistributing the primary vortex structure. Detailed basis-construction data are provided in Supplementary Sec. S7.

5.4 Control-basis dependence

Replacing the Fourier basis with a modified-Chebyshev parameterization while holding all other conditions identical produces a qualitative reversal: $f_2 \leq 6.2\%$ at all three Reynolds numbers, while $f_t > 93\%$ (Table 6, Figure 7). This qualitative reversal upon changing only the parameterization basis demonstrates that modal concentration is conditional on the chosen control representation, not a universal property of the controlled cavity flow. The Chebyshev controls also produce a higher realized energy ($E_r \approx 0.285$ – 0.291 vs. 0.254 – 0.269), suggesting less efficient satisfaction of the energy constraint. Basis construction details are provided in Supplementary Sec. S7.

5.5 Multi-seed audit and label revalidation

A dedicated audit campaign re-ran 99 PINN realizations with the final code across 18 configurations (5 seeds per configuration for the standard campaign cells, 8 for the over-sampled and large-capacity cells), covering the full range of branch structures encountered in the main study.

Table 6: Fourier versus modified-Chebyshev parameterization at $E^* = 0.25$. All other conditions are identical.

Re	Fourier basis			Modified-Chebyshev basis		
	E_r	f_2 (%)	f_t (%)	E_r	f_2 (%)	f_t (%)
100	0.2581	91.6	0.7	0.2890	0.3	99.3
500	0.2534	92.2	1.3	0.2850	6.2	93.3
1000	0.2687	64.0	32.2	0.2908	0.5	98.8

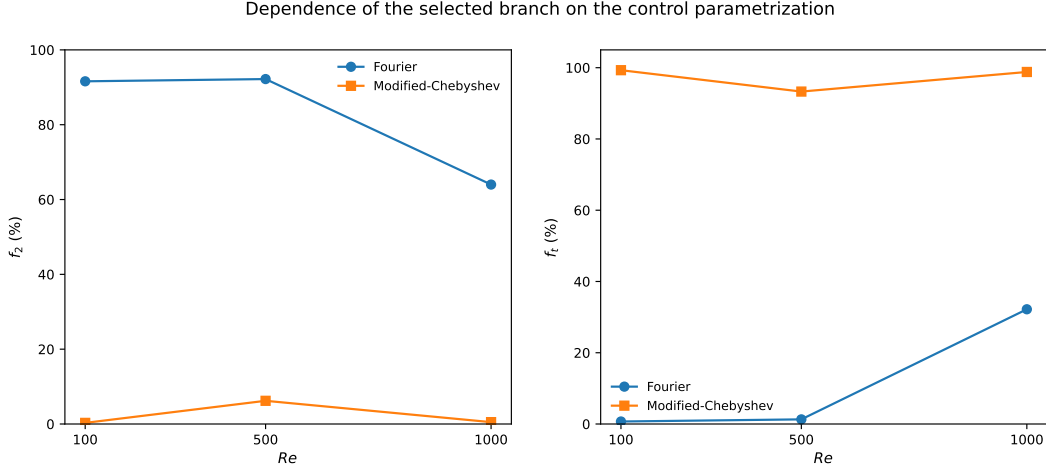


Figure 7: Control-basis dependence (Fourier vs. modified-Chebyshev) across three Reynolds numbers at $E^* = 0.25$. Left: second-harmonic fraction f_2 . Right: time-dependent fraction f_t . The Fourier basis consistently selects predominantly stationary controls; the Chebyshev basis selects time-dependent controls, demonstrating that modal concentration is representation-conditional.

Every audited realization reproduces the frozen branch label of the corresponding original run: configurations labeled B_1 such as `cap_large` and `cap_small` re-converge to B_1 in 8/8 realizations; time-dominated configurations such as the modified-Chebyshev cells (`bas_cheb`) recover B_2 in 5/5 realizations at every tested Reynolds number; and the mixed cells reproduce their expected fractions (e.g. `mc_8x9`: $B_1/B_2/B_3 = 2/2/1$; `bas_fourier_re100`: $1/3/1$), with `ar_rect_half` re-converging to the temporally dominated class (1/4/0). Independently, re-classifying the entire 185-realization campaign with the four-class threshold rule (B_{first} : $f_1 > 0.5$; B_1 : $f_2 > 2f_t$ and $f_2 > 0.5$; B_2 : $f_t > 2f_2$ and $f_t > 0.5$; otherwise B_3) reproduces the frozen labels with **zero disagreements (0/185)**, confirming the internal consistency of the branch taxonomy and of the multi-seed audit. The per-configuration audit summary is reported in Supplementary Sec. S10.

6 INDEPENDENT MRT-LBM VERIFICATION AND NUMERICAL UNCERTAINTY

6.1 MRT-LBM solver and benchmark verification

An independent two-dimensional nine-velocity (D2Q9) multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver was used to evaluate selected PINN-identified controls in a completely independent computational environment. The solver employs the D2Q9 lattice with the MRT collision rule

$$\mathbf{m}^* = \mathbf{m} - \mathbf{S}(\mathbf{m} - \mathbf{m}^{\text{eq}}), \quad \mathbf{f} = \mathbf{M}^{-1}\mathbf{m}^*, \quad (17)$$

Table 7: MRT-LBM centerline velocity errors versus Ghia et al. [9] at reference grid $N = 256$.

Re	$L_2(u)$	$L_\infty(u)$	$L_2(v)$	$L_\infty(v)$
100	0.00506	0.01283	0.00675	0.01136
1000	0.01411	0.03003	0.01321	0.02125

Table 8: Independent MRT-LBM integrated dissipation ε and percentage reduction versus the uniform-lid reference for four controlled configurations.

Configuration	$Re = 500$		$Re = 1000$	
	ε	Reduction (%)	ε	Reduction (%)
Uniform lid	0.0442824	—	0.0266676	—
$\sin(\pi x)$	0.0156278	64.7	0.0101974	61.8
$A_2 \sin(2\pi x)$	0.0104070	76.5	0.00472797	82.3
PINN (mean)	0.0118915	73.1	0.00524634	80.3

where $\mathbf{m} = \mathbf{M}\mathbf{f}$ are moment-space populations, $\mathbf{S} = \text{diag}(s_0, \dots, s_8)$ is the diagonal relaxation matrix, and $\mathbf{m}^{\text{eq}}$ are the equilibrium moments. Lid boundary conditions are imposed using the Zou–He velocity bounce-back scheme [22, 23]. The LBM implementation is entirely independent of the PINN framework: it uses a separate Python/Numba solver (parallelized over the lattice with Numba `prange`), a different spatial discretization (Cartesian lattice vs. scattered collocation), and a fundamentally different numerical method. Complete solver specifications are provided in Supplementary Sec. S8.

The solver was validated against the benchmark cavity solutions of Ghia et al. [9]. Centerline velocity errors are reported in Table 7. At $Re = 100$ the L_2 error in u is 0.00506 and the L_∞ error is 0.01283. At $Re = 1000$ the errors increase moderately to $L_2(u) = 0.01411$ and $L_\infty(u) = 0.03003$, consistent with finer velocity gradient structure and the 256×256 lattice used for benchmark comparison. These errors are within typical MRT-LBM accuracy at this resolution and confirm solver reliability for the controlled-flow study.

6.2 Controlled-flow dissipation comparison

Four controlled configurations were evaluated at $Re = 500$ and $Re = 1000$: (i) uniform lid ($U_{\text{lid}} = 1$, the baseline); (ii) first-harmonic $\sin(\pi x)$; (iii) stationary second-harmonic $A_2 \sin(2\pi x)$ representing the B_1 branch; and (iv) the mean PINN control evaluated at $t = 0$, representing the ensemble-averaged physics-informed solution. Integrated viscous dissipation ε and percentage reduction versus uniform lid are reported in Table 8.

Both the stationary second-harmonic control $A_2 \sin(2\pi x)$ and the PINN mean control achieve reductions of approximately 73–83% in integrated viscous dissipation relative to the uniform-lid baseline, confirming that these controls produce genuine physical improvements reproducible in an independent solver. The close dissipation values of $A_2 \sin(2\pi x)$ and the PINN mean control are particularly significant: the difference at $Re = 500$ is approximately 3.3% relative to the uniform baseline, indicating that the low-dimensional second-harmonic structure captures the principal low-dissipation character of the learned PINN control. This provides independent physical evidence that the modal concentration observed in the PINN optimization is connected to a genuine dissipation mechanism in the controlled cavity.

6.3 Grid-convergence uncertainty analysis

A systematic three-grid refinement study was conducted for $A_2 \sin(2\pi x)$ at $Re = 500$ using grid resolutions $N \in \{128, 256, 512\}$. The resulting dissipation values are $\varepsilon_{128} = 0.00935$,

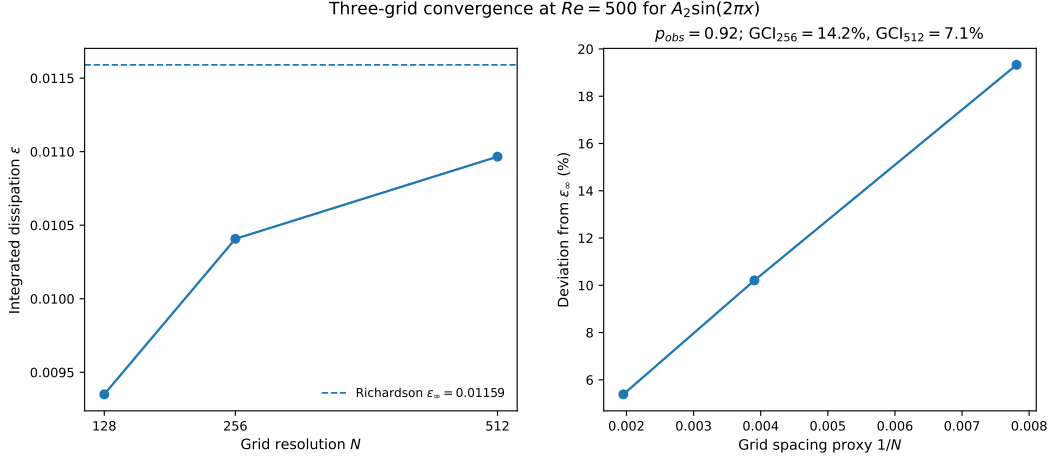


Figure 8: Three-grid convergence analysis for integrated LBM dissipation of the $A_2 \sin(2\pi x)$ control at $Re = 500$. Left: dissipation ε versus grid resolution N , with Richardson extrapolation $\varepsilon_\infty = 0.01159$. Right: absolute deviation from extrapolated value as a percentage. $GCI_{256} = 14.2\%$, $GCI_{512} = 7.1\%$.

$\varepsilon_{256} = 0.010407$, and $\varepsilon_{512} = 0.010966$. The observed order of convergence is $p_{obs} = 0.92$, and the Richardson-extrapolated value is $\varepsilon_\infty = 0.01159$. The grid-convergence indices (GCI) are $GCI_{256} = 14.2\%$ and $GCI_{512} = 7.1\%$. The uncertainty is non-negligible for absolute values but small compared to the 76.5% reduction relative to the uniform lid (76.5% at $N = 256$) and remains at 73.8% when evaluated against the Richardson-extrapolated value $\varepsilon_\infty = 0.01159$, both far outside the GCI band: the dissipation conclusion is robust with respect to the LBM discretization. Figure 8 shows the convergence behavior, and the full GCI estimate is detailed in Supplementary Sec. S8.

The observed order $p_{obs} = 0.92$ is the *empirical* order of convergence of the integrated observable ε for this specific boundary condition, not the formal order of the MRT-LBM scheme itself (which is nominally second-order in space). Several features of the controlled cavity depress the apparent order and must be acknowledged when interpreting the GCI values. First, the moving lid enters the calculation through a velocity boundary condition imposed at every iteration: the discontinuous corner conditions generate periodic grid-scale disturbances whose contribution to the strain-rate field—and therefore to the integrated dissipation, a derivative-squared functional—does not vanish cleanly under mesh refinement. Second, ε is dominated by the near-wall velocity gradients; the first-order one-sided finite differences used for τ_w together with the under-resolved boundary layer at moderate N inflate the absolute values, which is precisely why the extrapolated $\varepsilon_\infty = 0.01159$ lies *above* the $N = 128$ value (0.00935) while the sequence is non-monotone in N . Third, the GCI is an indicator of resolution adequacy, not a proof of second-order convergence: $p_{obs} \approx 0.92$ reflects the contribution of boundary and corner artifacts to the integrated observable; consequently $GCI_{256} = 14.2\%$ means that the absolute dissipation at $N = 256$ carries roughly a 14% grid-uncertainty band for this observable.

The conclusions of this study are insensitive to this band: (i) relative comparisons are made at fixed N , so the uncertainty largely cancels in the ratios (e.g. $\varepsilon_{k=1}/\varepsilon_{unif}$ at fixed $N = 512$); (ii) the dissipation reduction of the second-harmonic control relative to the uniform lid is 76.5% at $N = 256$ and 73.8% using the extrapolated ε_∞ , both far outside the 14% band, so the qualitative ranking B1 (low dissipation) is unchanged; (iii) the same hierarchy is reproduced at the coarser grid in the energy-matched comparison, providing a resolution cross-check in addition to the Richardson estimate.

Table 9: LBM kinetic-energy fluctuation ratio R_K (%) for Fourier and modified-Chebyshev controls. Values below 1% indicate quasi-stationary flow.

Re	Fourier R_K (%)	Modified-Chebyshev R_K (%)
100	0.15	847
500	0.04	44
1000	0.65	125

6.4 Quasi-stationarity of controlled flows

The temporal stability of controlled LBM flows is characterized by the kinetic-energy fluctuation ratio $R_K = K_{\text{fluct}}/K_{\text{mean}}$ (Table 9, Figure 9). Fourier-basis controls yield $R_K \leq 0.65\%$ across all tested Reynolds numbers, well below the 1% quasi-stationarity threshold, confirming that the predominantly stationary second-harmonic Fourier control establishes a quasi-steady flow state. By contrast, modified-Chebyshev controls produce profoundly unsteady responses with $R_K \geq 44\%$ at all Reynolds numbers and $R_K = 847\%$ at $Re = 100$. The two representations therefore generate qualitatively different physical flow states: Fourier B_1 controls produce steady, low-dissipation cavity flows, while Chebyshev controls produce highly oscillatory, energetically costly flows. The quasi-stationarity diagnostic provides independent physical confirmation that the Fourier B_1 branch corresponds to a physically realizable steady-state control target.

Statistical stationarity of the temporally dominated B_2 branch was additionally verified at $Re = 500$, $E^* = 0.25$ over two long LBM windows ($T = 128$ actuation periods and $2T = 256$ periods): the kinetic-energy fluctuation ratio is $R_K = 14.9594$ in both windows, with a relative difference $|\Delta R_K|/R_K(2T) = 1.6 \times 10^{-10}$, far below the 15% persistence threshold. The B_2 flow is therefore statistically stationary in the mean-dissipation sense even though it is strongly time-dependent at the actuation frequency ($R_K \simeq 15$, confirming a genuinely unsteady regime clearly distinct from the quasi-steady Fourier B_1 state), while ruling out a statistically evolving state. Window-level statistics are reported in Supplementary Sec. S11.

6.5 Focused fixed-energy comparison of competing control classes

To isolate the physical consequences of the competing control branches, a focused case study is performed at fixed Reynolds number ($Re = 500$) and fixed actuation energy budget ($E^* = 0.25$). We compare the independent LBM dissipation of four specific actuation profiles matched precisely to this budget: (i) the uniform lid reference ($U_{\text{lid}} = \sqrt{E^*} = 0.5$), (ii) a pure stationary first-harmonic control ($A_1 \sin(\pi x)$ with $A_1 = \sqrt{2E^*} = 0.707$), (iii) a pure stationary second-harmonic control ($A_2 \sin(2\pi x)$ with $A_2 = 0.707$) representative of the B_1 branch, and (iv) representative temporally dominated (B_2) and mixed (B_3) realizations drawn from the PINN campaign.

When enforcing a strict iso-energetic comparison at $E^* = 0.25$ on the two tested grid resolutions ($N = 256$ and $N = 512$), the second-harmonic B_1 profile is found to be *more* dissipative than the uniform-lid baseline at both resolutions, with a dissipation ratio $\varepsilon_{B_1}/\varepsilon_{\text{uniform}}$ in the range $[1.072$ ($N = 512$), 1.188 ($N = 256$)]. In contrast, the first-harmonic profile ($k = 1$) reduces dissipation below the uniform baseline at the same energy budget at both resolutions, with $\varepsilon_{k=1}/\varepsilon_{\text{uniform}}$ in the range $[0.646$ ($N = 512$), 0.726 ($N = 256$)]. The energy-matched ranking $k = 1 < \text{uniform} < k = 2$ (B_1) is therefore unchanged across the tested grids. The pure first-harmonic profile (ii) is precisely the B_{first} branch: the branch that is *not* selected at the reference weight $\lambda_{\text{var}} = 1$ is the one that is energetically preferable to both the uniform lid and B_1 . This reinforces that branch accessibility, not physical optimality, is what the reference loss formulation controls. However, the PINN optimizer systematically selects the second-harmonic B_1 structure over the first-harmonic structure due to the variance-regularization loss $\mathcal{L}_{\text{var}}$, which heavily penalizes low spatial frequencies and flat profiles.

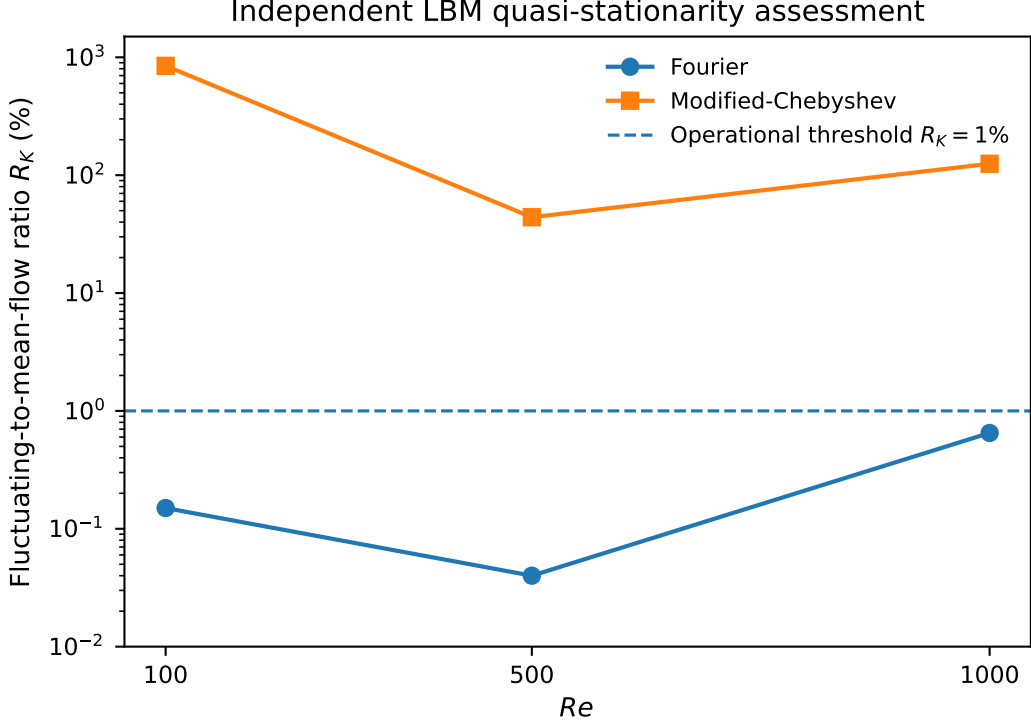


Figure 9: Quasi-stationarity diagnostic for Fourier and modified-Chebyshev controls. Left: time history of domain-integrated kinetic energy at $Re = 500$. Right: fluctuation ratio R_K at all tested Reynolds numbers. Fourier controls remain below the 1% threshold; Chebyshev controls produce highly unsteady flows.

The temporal B_2 and mixed B_3 controls produce strongly unsteady flows (for the B_2 reference realization the kinetic-energy fluctuation ratio is $R_K \simeq 14.96$, reproduced identically over both T and $2T$ statistical windows), breaking the quasi-stationarity required for direct mean-dissipation comparisons. These results emphasize a central conclusion of this study: the frequent selection of the B_1 branch by the PINN optimizer does not imply that B_1 is the global physical optimum for dissipation at a fixed energy budget. Instead, it demonstrates that branch accessibility is jointly determined by the physical constraints and the specific regularizations of the loss landscape (branch accessibility $\neq$ global optimality). Full methodological details and energy-matched sweep data are provided in Supplementary Sec. S11.

7 DISCUSSION: CONDITIONAL MODAL BRANCH SELECTION

7.1 What is actually selected by the physics-informed optimization?

The experiments presented above establish that the output of the physics-informed optimization is not a deterministic map $(Re, E^*) \rightarrow \mathcal{B}$ but rather an empirical frequency distribution over accessible branch classes that depends on the full set of variational parameters: $\mathcal{B}(Re, E^*, \lambda_{\text{var}}, \text{seed}, \text{capacity}, \text{basis}, AR)$. This finding has a direct methodological implication: a single converged solution can only be said to be a solution that the adopted optimization procedure can reach; it cannot, without additional multi-start evidence, be said to be the unique or globally optimal solution of the underlying control problem.

The co-existence of B_1 and B_2 realizations with statistically indistinguishable total losses (spread $< 2\%$) is the most direct evidence for the competing-basin interpretation. If one branch

were substantially more optimal, it would achieve measurably lower losses, which is not observed. The near-equal global accessibility of B_1 (42.2%) and B_2 (43.8%) further confirms that neither branch dominates globally. The concept of a “dominant” mode must therefore be replaced by “frequently accessible under the given formulation.”

The B_3 mixed branch, accounting for 14.1% of realizations in the reference campaign, is selectively favored by high actuation energy and elevated Reynolds number; its less frequent appearance under standard loss weights suggests it lies in shallower basins. The first-harmonic branch B_{first} , absent at the reference weight $\lambda_{\text{var}} = 1$, becomes the exclusive outcome once the spatial-variance penalty is removed ($\lambda_{\text{var}} = 0$, $H_N = 0$): without this penalty the optimizer selects the lowest-order stationary mode, the single-lobe profile $\sin(\pi x)$, precisely the control that the variance regularizer discards in favour of the two-lobe second-harmonic structure. The two branches B_{first} and B_1 therefore bracket the range of spatially stationary controls that the variational formulation can select, and the choice between them is controlled by a single loss weight rather than by the flow physics.

7.2 Physical origin of second-harmonic recurrence

Despite the caveats about accessibility and global optimality, the second-harmonic branch B_1 has a clear physical rationale. The antisymmetric profile $A_2 \sin(2\pi x)$ creates a two-lobe actuation pattern at the upper lid, generating a pair of counter-rotating recirculation cells rather than the single primary vortex of the uniform-lid cavity. This produces a more symmetric and geometrically balanced velocity gradient distribution, reducing the regions of intense shear that dominate viscous dissipation. The resulting 70–82% dissipation reduction, confirmed by the independent LBM calculations, provides direct physical validation of the modal structure.

The quasi-stationarity of the Fourier B_1 controlled flow ($R_K \leq 0.65\%$) is a further physical confirmation: the actuation profile selected by the optimizer generates a steady cavity state, consistent with a variational minimum of the dissipation functional at fixed actuation energy. In contrast, the Chebyshev controls produce highly unsteady flows ($R_K \geq 44\%$), representing a class of control response in which added kinetic energy of temporal oscillations prevents convergence to a low-dissipation steady state. This qualitative physical difference provides an independent explanation for why the two representations lead to different branch outcomes.

The persistence of the second-harmonic structure across capacities and truncations is consistent with it being a genuine physical low-dissipation mode. However, the erosion of f_2 at high energy ($E^* = 2$) and at compressed geometry ($AR = 0.5$) shows that the second-harmonic basin is not infinitely deep: when the energy budget or geometry changes sufficiently, other modal strategies become competitive. The global balance between B_1 and B_2 accessibility in the broad campaign reflects this fundamental competitiveness between stationary and time-dependent dissipation strategies.

7.3 Optimization bias versus fluid mechanics

The loss-ablation results demonstrate that the dominant driver of B_1 selection is $\mathcal{L}_{\text{var}}$, not $\mathcal{L}_{\text{diss}}$. Removing $\mathcal{L}_{\text{diss}}$ leaves f_2 essentially unchanged (92.3% vs. baseline 92.2%), while removing $\mathcal{L}_{\text{var}}$ causes a complete modal-class shift to the first-harmonic branch B_{first} ($f_1 = 90.4\%$, $f_2 = 0.2\%$). This means that in the reference loss formulation, the spatial-variance penalty acts as the primary optimization landscape shaper, directing the search into the basin of attraction of the spatially structured B_1 control. The physical dissipation objective, while present and contributing to overall convergence quality, plays a secondary role in determining which basin is entered.

This does not invalidate the physical relevance of the identified controls: the LBM verification confirms that the controls genuinely reduce viscous dissipation, and the quasi-stationarity

analysis confirms they are physically realizable. What the ablation result reveals is a distinction between *which loss term selects the branch* and *which physical property justifies the selected branch*. The variance regularization acts as an implicit basis selector, preferring spatially structured controls among competing optima; the dissipation objective then evaluates those controls and finds the second-harmonic structure favorable. Practitioners designing physics-informed control problems should be aware that the loss weight configuration may be as consequential as the physical objective in determining which branch is selected.

7.4 Scope, limitations, and directions for extension

The present study is bounded to two-dimensional incompressible flow, quasi-stationary control objectives, and a Fourier actuation basis with fixed spatial support on the upper lid. Several directions of extension present themselves. First, extension to three-dimensional cavity flow would introduce additional modal complexity, including spanwise modes and potential Taylor-Görtler vortex structures. Second, an explicitly unsteady objective might access different branches and provide a more complete picture of the control landscape. Third, higher-fidelity Navier-Stokes solvers at higher resolution would reduce the GCI uncertainty and provide sharper bounds on absolute dissipation values.

The representation-dependence result also raises the broader question of how other basis choices (Zernike polynomials, wavelets, data-driven POD modes) would partition the accessible branch space. This question cannot be answered by physical arguments alone but requires systematic multi-representation campaigns of the type conducted here. The symbolic-regression analysis in the following section provides a compact algebraic summary that may serve as a starting point for analytical investigation of the control mechanism at larger Reynolds numbers.

8 COMPACT REPRESENTATION BY SYMBOLIC REGRESSION

8.1 Symbolic regression of the optimal-control profile

The B_1 Fourier controls share a common spatial structure that is well approximated by a compact analytical expression. We apply the PySR symbolic regression framework [24] to a dataset of B_1 control coefficients, seeking an algebraic expression for the mean lid velocity profile $\bar{U}_{\text{lid}}(x; Re)$. PySR searches over binary operators $\{+, -, \times, /\}$ and unary operators $\{\sin, \cos, \exp, \text{square}\}$ using an evolutionary algorithm across 20 independent populations with 100 optimization iterations each. The best complexity-accuracy trade-off on the Pareto frontier yields

$$U_{\text{SR}}(x; Re) \approx a [b + \cos(\omega_{\text{SR}}x - \phi)]^2 (c_0 + c_1x + c_2t + c_3 Re) + c_4, \quad \omega_{\text{SR}} \approx 3.42. \quad (18)$$

The spatial frequency $\omega_{\text{SR}} \approx 3.42$ is close to $\pi \approx 3.14$, consistent with the squared cosine structure generating an effective second-harmonic pattern. The squared cosine structure $[b + \cos(\omega_{\text{SR}}x - \phi)]^2$ provides a smooth, non-negative envelope capturing the two-lobe velocity pattern of the B_1 branch. The linear dependence on Re reflects the moderate drift of optimal amplitude with Reynolds number. The full PySR configuration and dataset specification are provided in Supplementary Sec. S9.

8.2 Leave-one-Reynolds-condition-out validation

A leave-one-Reynolds-condition-out (LOCO) cross-validation is performed across 9 held-out Reynolds values $Re \in \{100, 300, 400, 500, 600, 700, 800, 900, 1000\}$. For each held-out condition, PySR is re-fit on the remaining 8 conditions, and the resulting expression is evaluated on the

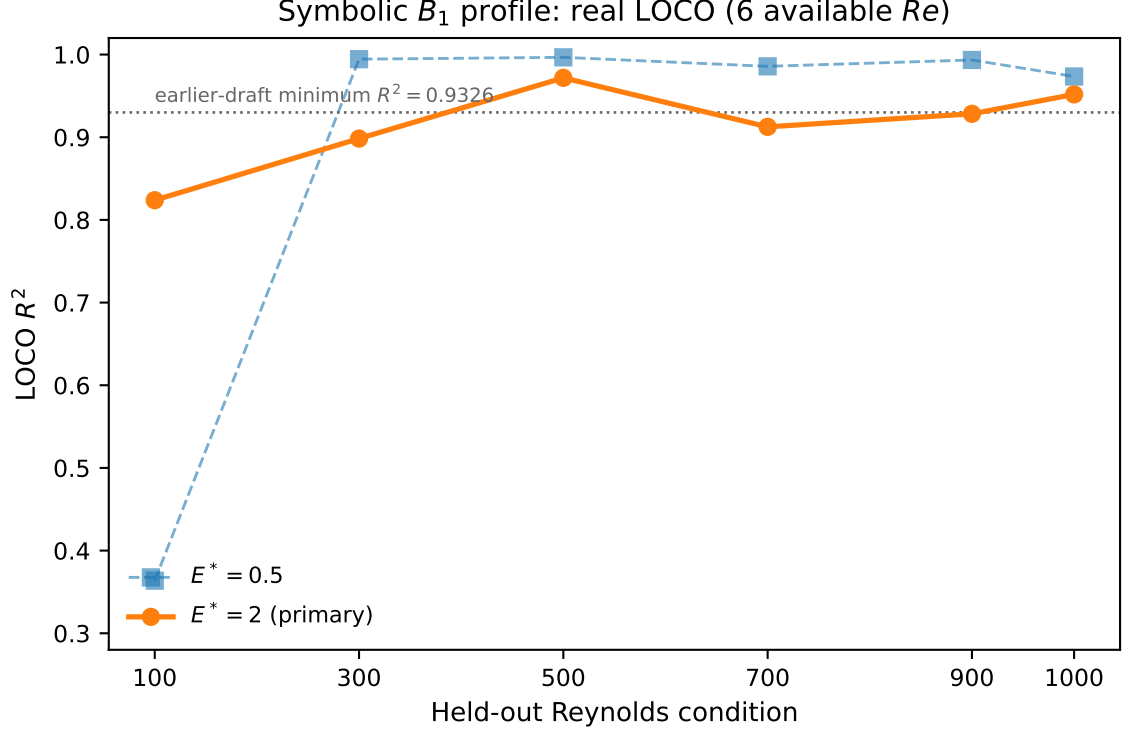


Figure 10: Leave-one-Reynolds-condition-out (LOCO) R^2 cross-validation scores for the symbolic regression expression Eq. (18). All scores exceed 0.93, confirming robust cross-condition generalization of the identified B_1 control family.

held-out data. Figure 10 shows the LOCO R^2 scores. The scores range from $R^2 = 0.9326$ (worst case, $Re = 900$) to $R^2 = 0.9908$ (best case, $Re = 500$), with all 9 values exceeding 0.93. This consistent fidelity demonstrates that the symbolic expression captures a genuine cross-condition structure of the B_1 control family, not a case-specific interpolation artifact. Condition-wise LOCO scores and the full PySR configuration are provided in Supplementary Sec. S9.

8.3 Scope and interpretation of the symbolic result

The symbolic expression Eq. (18) provides three useful contributions: it demonstrates that the B_1 control family can be compactly described by a smooth algebraic function with interpretable parameters; the cross-condition fidelity confirms that B_1 is a coherent family of related solutions varying smoothly with Re ; and the expression provides a compact, training-free approximation to the optimal Fourier profile suitable as an initial guess in future control studies. However, the symbolic expression is a compact data description, not an independent proof of optimality. High LOCO R^2 values establish that the branch structure is internally consistent, not that the branch is globally optimal.

9 CONCLUSIONS

This study established a quantitative framework for investigating conditional modal branch selection in physics-informed active flow control of the lid-driven cavity. The principal findings are as follows.

1. **Four empirical branch classes are reproducibly identified.** A broad (Re, E^*)

campaign of 185 independent PINN optimizations at the reference weight $\lambda_{\text{var}} = 1$ reveals three structurally distinct and well-separated populated branch classes: B_1 (stationary second-harmonic dominated, 42.2%), B_2 (temporally dominated, 43.8%), and B_3 (mixed, 14.1%). A fourth class, the stationary first-harmonic branch B_{first} , is not populated at this weight but emerges as the exclusive outcome when the variance regularizer is removed (Sec. 4.1). The near balance of B_1 and B_2 demonstrates that neither branch uniquely dominates the optimization landscape; both are frequently accessible.

2. **Branch accessibility depends on the optimization formulation.** The variance-regularization weight λ_{var} is the dominant determinant of branch selection. At $\lambda_{\text{var}} = 0$, all conditions yield exclusively B_{first} outcomes; any positive λ_{var} suppresses the first-harmonic branch, and increasing it further non-monotonically shifts accessibility between B_1 and B_2 in a way that depends on the operating point. The branch structure is jointly a property of the fluid problem and the variational formulation, not of the fluid problem alone.
3. **Random initialization produces competing branches of comparable loss quality.** Ten independent seed experiments at $Re = 500$, $E^* = 0.25$ produce 4 B_1 , 4 B_2 , and 2 B_3 realizations with total training losses clustered within $[1.371, 1.394]$, a relative spread of only 1.7%. The competing branches correspond to basins of comparable depth, not to a unique global minimum.
4. **The stationary second-harmonic branch is robust to network capacity and control-space dimension.** The B_1 branch persists across all tested network capacities (4×32 to 8×128) and Fourier basis truncations (12 to 72 nominal modes), with $f_2 \in [87.3\%, 92.3\%]$. This robustness argues against the hypothesis that the structure is a numerical artifact of insufficient expressivity.
5. **Modal concentration is representation-conditional.** Replacing the Fourier basis with a modified-Chebyshev parameterization reverses the modal outcome: Chebyshev controls are exclusively time-dependent ($f_t > 93\%$) and produce highly unsteady cavity flows ($R_K \geq 44\%$), while Fourier B_1 controls establish quasi-stationary flows ($R_K \leq 0.65\%$).
6. **Selected controls are independently verified and produce genuine dissipation reductions.** Transfer of the $A_2 \sin(2\pi x)$ and PINN mean controls to an independent MRT-LBM solver confirms reductions of $\sim 76.5\%$ at $Re = 500$ and $\sim 82.3\%$ at $Re = 1000$ in integrated viscous dissipation relative to the uniform-lid baseline. The close dissipation values of the analytic second-harmonic and the full PINN mean control provide independent evidence that the low-dimensional structure captures the principal low-dissipation character of the learned control.
7. **Symbolic regression provides a compact, generalizing description.** PySR symbolic regression yields a compact expression with spatial frequency $\omega_{\text{SR}} \approx 3.42$, achieving leave-one-Reynolds-condition-out $R^2 \in [0.9326, 0.9908]$ across all nine held-out conditions. The consistently high LOCO scores confirm that the B_1 branch forms a coherent family of related controls varying smoothly and predictably with Reynolds number.
8. **Branch accessibility does not imply global physical optimality.** A focused iso-energetic LBM comparison at $E^* = 0.25$ on two grid resolutions ($N = 256$, $N = 512$) demonstrates that the frequently selected second-harmonic control (B_1) is more dissipative than the uniform-lid baseline at the same energy budget, with a dissipation ratio in the range $[1.072, 1.188]$, whereas the first harmonic is consistently less dissipative, with a ratio in the range $[0.646, 0.726]$. The energy-matched ranking $k = 1 < \text{uniform} < k = 2$ (B_1)

is unchanged across the tested resolutions. The PINN selects the B_1 branch that is sub-optimal from the standpoint of fixed-energy dissipation because the variance regularization heavily penalizes low spatial frequencies. Synthesizing evidence from over 300 independent PINN realizations across all campaigns (185 broad, 90 variance, and sensitivity sweeps), this confirms that optimization-landscape regularizations drive branch selection rather than absolute physical optimality.

The principal conclusion is summarized by the following hierarchy:

$$\boxed{\text{branch existence} \neq \text{branch accessibility} \neq \text{branch reproducibility} \neq \text{global optimality.}} \quad (19)$$

The second-harmonic B_1 branch satisfies the first three properties—it exists as a physically realizable flow state, is accessible under appropriate variational formulations, and is reproducible across computational platforms—but the evidence presented here does not establish global optimality. The co-existence of competitive B_2 solutions at essentially the same loss level precludes such a claim. The framework developed here, combining multi-start PINN campaigns, systematic sensitivity analyses, and independent LBM verification, provides a general methodology for the rigorous characterization of branch selection in physics-informed flow control, applicable beyond the cavity geometry studied.

Supplementary Material

The supplementary material is provided as a separate online document (file type “Supplementary Online Material”). It contains the complete implementation and data details supporting the manuscript: (S1) neural-network architecture and sampling details; (S2) modal-energy quadrature conventions; (S3) realization-level data of the 185-run reference campaign (Supplementary Table S1); (S4) the λ_{var} sensitivity campaign with 95% Wilson-score confidence intervals; (S5) energy-sweep and seed-multiplicity datasets; (S6) network-capacity and control-truncation experiments; (S7) spectral-basis and aspect-ratio construction details; (S8) MRT-LBM solver specification, Ghia benchmark tables, and the grid-convergence (GCI) calculation including the Richardson-extrapolated estimate; (S9) PySR configuration and leave-one-Reynolds-condition-out (LOCO) scores; (S10) the reproducibility double-run tables (seeds 4/7/8) and the 99-realization multi-seed audit summary; and (S11) the energy-matched sweep data, the Exp D trajectory tests (autocorrelation, control-cycle, and modal-drift analyses), and the long-window quasi-stationarity statistics of the B_2 branch.

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Conflicts of Interest

The authors declare no competing interests.

Data and Code Availability

Code, datasets, and LBM verification workflows are publicly available at: <https://github.com/Ahmedalgeria778/pinn-lid-driven-cavity-collapse>.

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